Multi-fidelity Bayesian optimization of mixing in two-phase rocking bioreactors

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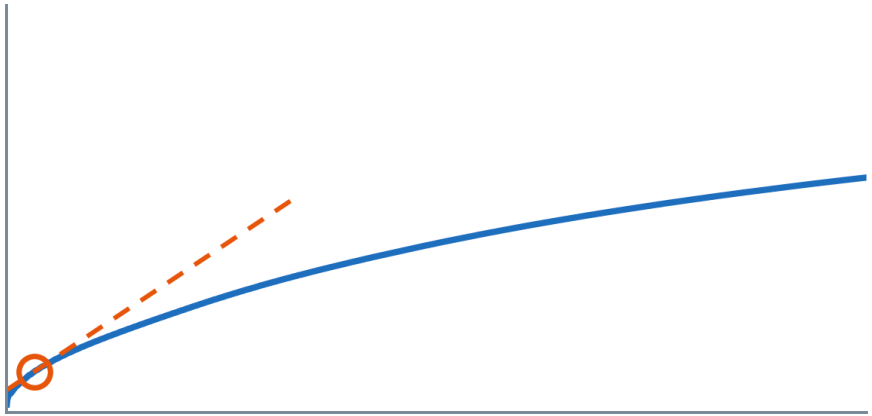


Motivation

Introduction

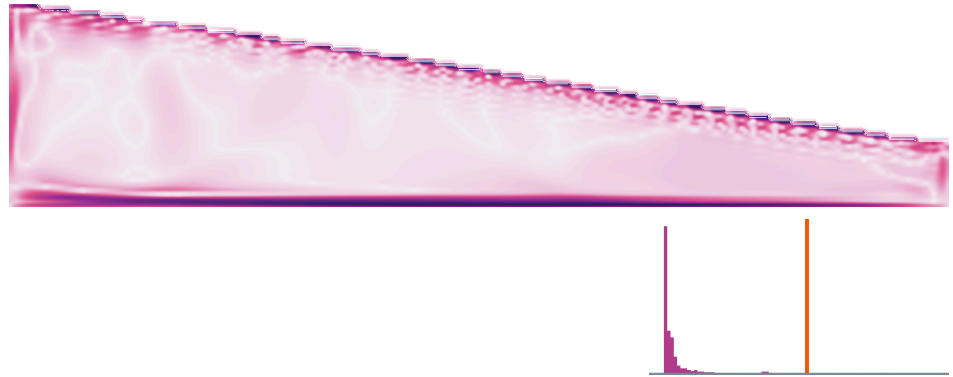
kLa

$$dC^*/dt = kLa\ (1 - C^*)$$



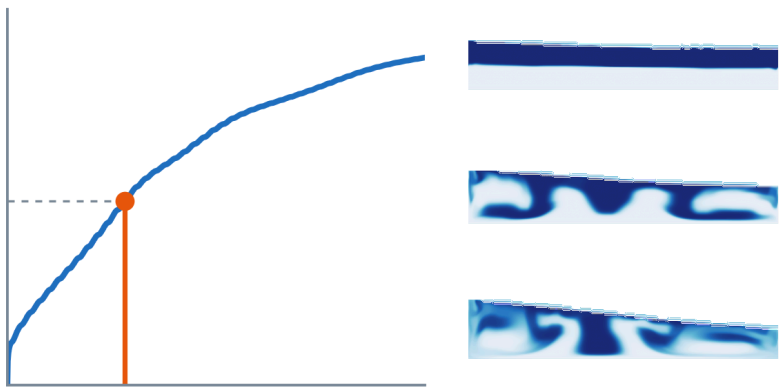
shear stress

$$\tau = \mu\ (\partial u/\partial y + \partial v/\partial x)$$



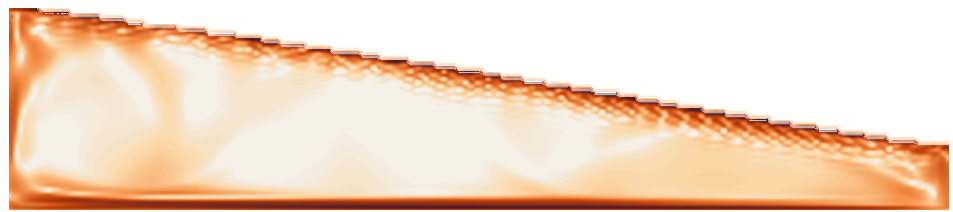
mixing time

$$\chi = 1 - \sigma^2 / \sigma^2 max$$



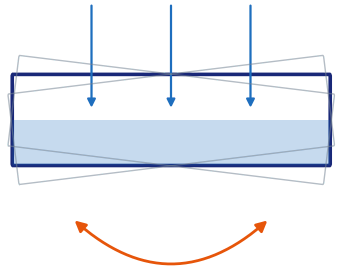
dissipation rate

$$\varepsilon = 2\ \mu\ S : S$$



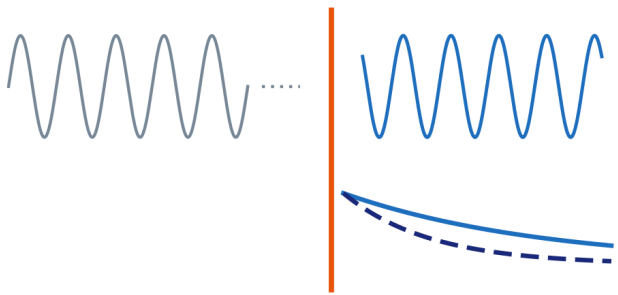
Basilisk

Navier–Stokes + VOF

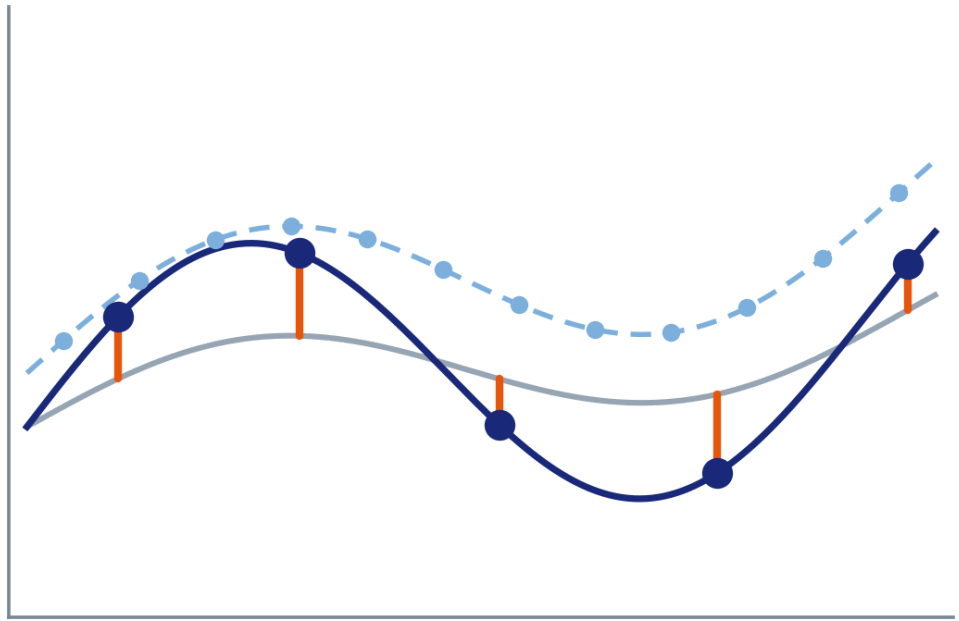


release

$$\partial c/\partial t + u\cdot\nabla c = \nabla\cdot(D\nabla c)$$



multi-fidelity



$$f_H(x) = \rho\, f_L(x) + \delta(x)$$

$\delta \sim \mathrm{GP}$

Results 1

Results 2

Conclusions

Questions?